
Revisiting ‘Little Jiffy, Mark IV’*

Towards a Bayesian KMO index

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Abstract

The *Kaiser-Meyer-Olkin* (KMO) *index* is a measure of sampling adequacy used by researchers to assess whether a data matrix is factorable prior to a factor analysis. Since its conception, the KMO index has remained a frequentist statistic, leaving researchers unable to employ the advantages of Bayesian inference when assessing sampling adequacy. Building on the increasing relevance of the Bayesian statistical approach, as well as advancements in Markov-Chain Monte Carlo methods, the author proposes a reconceptualization of the KMO index within the Bayesian framework that enables researchers to incorporate prior information and make coherent probabilistic statements about the sampling adequacy of a data matrix.©¹

Keywords: *Kaiser-Meyer-Olkin index, KMO index, Bayesian Kaiser-Meyer-Olkin index, BKMO index, measure of sampling adequacy, MSA, Bayesian measure of sampling adequacy, BMSA, robust BKMO index, bootstrap KMO index, Bayesian inference, frequentist inference*

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1 Introduction

The *Kaiser-Meyer-Olkin* (KMO) index, formally referred to as a *measure of sampling adequacy* (MSA; Kaiser, 1970, 1981; Kaiser and Rice, 1974), is a statistic named in honor of Henry F. Kaiser², Edward P. Meyer³, and Ingram Olkin⁴ (Dziuban and Shirkey, 1974) that is often used to assess whether a data matrix is *factorable*, which is a desirable property to ascertain prior to a factor analysis (Field, 2018: 778-833). While originally introduced as 'a second generation Little Jiffy'⁵ by Kaiser (1970), the currently most used iteration of the KMO index is the result of two subsequent revisions⁶, including a collaboration between Kaiser and John Rice⁷ (1974), titled 'Little Jiffy, Mark IV'.⁸ Though it historically may have seen most use within psychology (e.g., Dziuban and Shirkey, 1974; Zabihi et al., 2014; Ghadiri et al., 2015), it remains a standard measure in statistical software (e.g., IBM, 2025; Lüdecke et al., 2021; Revelle, 2025) that helps researchers across disciplines gauge whether their data can be meaningfully explored using factor analysis (e.g., Ahmad et al., 2024; Almeida et al., 2022; Danaei et al., 2024; Sadowska et al., 2021; Sarfaraz et al., 2024).

While popular, the KMO index can be shown to conceptually belong to the *frequentist statistical framework* (for an introduction, see, e.g., Agresti, 2018), and a counterpart suitable for the alternative *Bayesian statistical framework* (for an introduction, see, e.g., McElreath, 2019) is currently missing. This is a relevant shortcoming, because these frameworks fundamentally differ in their conceptualization of probability and statistical uncertainty (Kruschke, 2014). Given that recent advances in the latter have made it more viable for applied research (Gelman, 2022), Bayesian versions of frequentist statistics are increasingly being developed (e.g., Ben-Shachar et al., 2020; Gelman et al., 2019; Makowski et al., 2019a). However, a Bayesian KMO index is currently missing, leaving researchers unable to exploit the numerous advantages of Bayesian inference, such as the ability to incorporate prior information into the analysis and make coherent probabilistic statements from a posterior distribution (Kruschke, 2014; McElreath, 2019).

This paper solves the issue of a missing Bayesian version of the KMO index by reconceptualizing it within the Bayesian framework and demonstrating its implementation using standard statistical software and empirical data. To build towards a Bayesian concept, this is achieved by revisiting the current KMO index (Section 2), reviewing its mathematical formula and demonstrating its frequentist conceptualization. The resulting *Bayesian Kaiser-Meyer-Olkin* (BKMO) index is then introduced (Section 3), being a *Bayesian measure of sampling adequacy* (BSMA) that can incorporate prior information and express uncertainty about the 'sampling adequacy' (i.e., factorability) of

²Former Professor of the University of California, Berkeley.

³Former professor of Loyola University, Chicago. It should be noted that there is some uncertainty about Edward P. Meyer being the 'Meyer' of 'Kaiser-Meyer-Olkin', because Henry F. Kaiser never appears to mention this 'Meyer' by first name, but only makes vague mentions of contributions from a 'Professor Meyer of Loyola, Chicago' (Kaiser, 1970: 405). At the same time, it seems that Kaiser did *not* himself coin the term 'Kaiser-Meyer-Olkin' as a name for the MSA, which instead appears to be an invention of Dziuban and Shirkey (1974), but these authors also make no explicit mention of a 'Professor Meyer'. However, based on personal communication with staff of Loyola University, Chicago, this 'Professor Meyer of Loyola, Chicago' is likely to be Edward Paul Meyer, who was a Professor of Psychology at Loyola during the period (1969 - 1972) when the KMO index was invented.

⁴Former Professor of Stanford University.

⁵The term 'Little Jiffy' was coined by former Professor Chester W. Harris of the University of Wisconsin-Madison and was used by Henry F. Kaiser to denote the analytic procedure of 'principal components with associated eigenvalues greater than one followed by normal varimax rotation' (cf. Kaiser, 1970: 402). Note that this 'Little Jiffy, Mark I' is *not* directly related to later versions of 'Little Jiffy' (e.g., Kaiser, 1970).

⁶The first revision of the KMO index, 'Little Jiffy, Mark III', went unpublished though its changes were incorporated into 'Mark IV' (Kaiser and Rice, 1974: 112).

⁷Emeritus Professor of the University of California, Berkeley.

⁸A final revision of the KMO index was made by Kaiser (1981), which is simply the square-root of the original KMO index by Kaiser (1970), but this revised version appears mostly ignored across statistical software packages and applied research. Accordingly, this paper focuses on 'Little Jiffy, Mark IV' (Kaiser and Rice, 1974).

a data matrix in the form of posterior distributions. Systematic comparisons using simulated data are conducted, which reveal that with both 'liberal' and 'conservative' priors, the BKMO index approximates the KMO index at even small samples, suggesting that their difference is primarily conceptual. This is followed by a detailed demonstration that shows how researchers can implement the BKMO index in relation to empirical data (Section 4), enabling inferential statements to be made regarding the 'sampling adequacy' given the data. The paper then discusses the developed BKMO index (Section 5), reviewing its areas of application and robustness, before a brief summary concludes this paper (Section 6).

2 The Kaiser-Meyer-Olkin index

To build towards a Bayesian reconceptualization, it is necessary to first understand the classical and most widely used version of the KMO index, that is, 'Little Jiffy, Mark IV' (Kaiser and Rice, 1974). This includes revisiting its mathematical formula and interpretation, as well as demonstrating its frequentist conceptualization.

Readers should note that this paper assumes familiarity with formula for standard statistical concepts (e.g., a correlation) and uses the terms 'KMO index', 'sampling adequacy', and 'factorability' interchangeably, since the KMO index *is* a measure of the factorability of a data matrix (cf. Kaiser, 1970, 1981; Kaiser and Rice, 1974). That notion also helps emphasize that the KMO index is dependent on the characteristics of the *data matrix* from which it is calculated.

Important information for understanding the formula involved in calculating the KMO index is revisited (Section 2.1), which serves to ease its interpretation (Section 2.2) and demonstrate its frequentist conceptualization (Section 2.3). The considerations made throughout this review will be used in the following section to derive a Bayesian reconceptualization.

2.1 Formula

To understand the formula underlying the KMO index, suppose first the existence of an n times k data matrix (i.e., $\mathcal{D}$), randomly sampled to a sufficient extent from a population of interest, where n is an integer denoting the number of rows, indexed by u , while k is an integer denoting the number of columns, indexed by m , with each column being a numeric vector. Assume further that each column has a finite mean and variance, and that the data matrix is *full column rank* (cf. Stock and Watson, 2019: 751). This data matrix is illustrated in definition (2.1.1).

$$\mathcal{D} \equiv \begin{bmatrix} \theta_{1,1} & \cdots & \theta_{1,k} \\ \vdots & \ddots & \vdots \\ \theta_{n,1} & \cdots & \theta_{n,k} \end{bmatrix} \quad (2.1.1)$$

With this general data matrix ($\mathcal{D}$) in mind, the mathematical formula typically used to calculate the KMO index can be shown. Recall the following standard statistical formula (Stock and Watson, 2019; Wooldridge, 2019), expressed in relation to $\mathcal{D}$:

$$\hat{\mu}_m \equiv \frac{1}{n} \sum_{u=1}^n \theta_{u,m} \quad (2.2.1)$$

$$\hat{\rho}_{m,m'} \equiv \frac{\sum_{u=1}^n (\theta_{u,m} - \hat{\mu}_m)(\theta_{u,m'} - \hat{\mu}_{m'})}{\sqrt{\sum_{u=1}^n (\theta_{u,m} - \hat{\mu}_m)^2} \sqrt{\sum_{u=1}^n (\theta_{u,m'} - \hat{\mu}_{m'})^2}} \quad (2.2.2)$$

where the defining equation (2.2.1) is the estimator for the *arithmetic mean*, and (2.2.2) is the *product-moment correlation coefficient* (Bravais, 1844; Pearson, 1895), with Σ denoting the summation operation, and the radical (i.e., $\sqrt{}$) denoting the square-root operation.

Finally, recall the concept of the *anti-image correlation coefficient* (Kaiser and Rice, 1974: 113-114), which for those unfamiliar is essentially the *partial* correlation coefficient (cf. Field, 2018: 358-361). It involves the correlation matrix (i.e., $\hat{P}$) and inverse correlation matrix (i.e., $\hat{P}^{-1}$), both indexed here by i and j , and it is specifically calculated from the individual correlation coefficients ($\hat{\rho}$) and inverse correlation coefficients (i.e., $\hat{\rho}^{-1}$) using formula (2.2.3), where $i \neq j$.

$$\hat{\rho}_{q[i,j]} \equiv -\frac{\hat{\rho}_{i,j}^{-1}}{\sqrt{\hat{\rho}_{i,i}^{-1} \hat{\rho}_{j,j}^{-1}}} \quad (2.2.3)$$

With these formula in mind, calculating the KMO index can then be done by letting $\text{KMO}(\mathcal{D})$ denote a function, defined in (2.2.4), for calculating the overall 'sampling adequacy' of a data matrix (i.e., $\hat{\alpha}$; Kaiser and Rice, 1974: 112-113). Similarly, calculating the individual KMO indices, that is, the MSA for item m in the data ($\mathcal{D}$), is achieved with function (2.2.5):

$$\text{KMO}(\mathcal{D}) \equiv \hat{\alpha} \equiv \frac{\sum_{i \neq j} \hat{\rho}_{i,j}^2}{\sum_{i \neq j} \hat{\rho}_{i,j}^2 + \sum_{i \neq j} \hat{\rho}_{q[i,j]}^2} \quad (2.2.4)$$

$$\text{KMO}(\mathcal{D}_m) \equiv \hat{\alpha}_m \equiv \frac{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2}{\sum_{j,j \neq i} \hat{\rho}_{i,j}^2 + \sum_{j,j \neq i} \hat{\rho}_{q[i,j]}^2} \quad (2.2.5)$$

where $\sum \sum$ denotes the row-wise summation operation, which is applied across the off-diagonal i and j indices of the k by k correlation matrix ($\hat{P}$) and anti-image correlation matrix (i.e., $\hat{P}_q$).

2.2 Interpretation

Based on these formula, the 'sampling adequacy' measured by the KMO index can be recognized as the ratio of the sum of squared inter-correlations relative to the sum of the squared inter-correlations and squared anti-image inter-correlations. A value close to 0 would indicate that 'the sum of [squared anti-image] correlations is large relative to the sum of [squared] correlations, indicating diffusion in the pattern of correlations' (Field, 2018: 798), while a value close to 1 would indicate that 'patterns of correlations are relatively compact and so factor analysis should yield distinct and reliable factors.' (Field, 2018: 798). Accordingly, the 'sampling adequacy' can more meaningfully be interpreted as the *factorability* of the data matrix, and for a factor analysis of the measured items to be appropriate, the 'sampling adequacy' should be as high as possible.

Given these formula, the KMO index value is generally a real scalar bounded between zero and one (i.e., $\text{KMO} \in \mathbb{R}^{(0,1)}$; cf. Kaiser and Rice, 1974: 113; Lüdtke et al., 2021), where a value of .5 occurs when the sum of squared inter-correlations are equal to the sum of the squared anti-image inter-correlations. Exceptions to this involve instances where the correlation matrix ($\hat{P}$) is composed of perfectly linearly independent variables (i.e.,

its off-diagonals are *zero vectors*; Boyd and Vandenberghe, 2018: 5), in which case the KMO index is undefined due to formula (2.2.4) then involving a division by zero. Similarly, if the correlation matrix is singular, the KMO index is *undefined* for the real number system because the inverse correlation coefficients cannot be derived for formula (2.2.3).

These exceptions can occur due to a lack of *sample variation* (cf. Wooldridge, 2019: 42) and *perfect multicollinearity* (cf. Wooldridge, 2019: 80), which is why the data matrix in section 2.1 was specified to eliminate such issues by being full column rank, with each variable having a finite variance. While these requirements are met by the specification of $\mathcal{D}$, they merely constitute assumptions for empirical data, and it is clear that they will also likely apply to a Bayesian reconceptualization.

2.3 Frequentist conceptualization

To corroborate the underlying claim of this paper, the classical KMO index can be shown to be frequentist. This is because of the fact that by merit of relying on the formula in section 2.1, the classical KMO index conceptually fails to (1) reflect statistical uncertainty by only providing a *point estimate* of 'sampling adequacy' and so require auxiliary processes (e.g., the estimation of standard errors), which contrasts Bayesian inference where all statistical uncertainty is inherently built into the posterior distribution (McElreath, 2019); (2) it cannot incorporate prior information, which is inherently non-Bayesian and means that information known *a priori* would be wasted in the estimation process; and (3) that it primarily serves to provide estimates of population parameters by treating data as *random* and the parameters as *fixed*, which is fundamentally different from the Bayesian approach, where it is the data that is treated as *fixed* (i.e., due to having been conditioned on), while the parameters are treated as *random* (cf. Kruschke, 2014: 125; McElreath, 2019: 10-11).

Consistent with Kruschke (2014), by merit of being frequentist, this means that the classical KMO index returns an estimate of the 'sampling adequacy' *in the population*, which is conceptually different from, and *not* necessarily equivalent to, a Bayesian KMO index that would instead be able to provide an estimate of the 'sampling adequacy' *given the data*.

Having considered the calculation and interpretation of the classical KMO index, as well as demonstrated its frequentist conceptualization, it is apparent that the classical KMO index is frequentist and in requirement of a reconceptualization to be compatible with Bayesian inference.

3 The Bayesian KMO index

This paper has so far revisited the classical KMO index, which involved reviewing the formula and interpretation of this measure of 'sampling adequacy' (Kaiser and Rice, 1974). That review demonstrated that the classical KMO index is conceptually frequentist, and the formula and concepts specific to that framework proved unable to estimate the posterior distribution and incorporate prior information, leaving it incompatible with Bayesian inference. To mitigate these limitations a *Bayesian Kaiser-Meyer-Olkin index* is introduced in this section.

This is accomplished by reconceptualizing the KMO index through use of Bayes' theorem and discussing the specification of priors and estimation of a posterior distribution using MCMC methods (Section 3.1). This results in an adjustment of the formula introduced in section 2.1 to derive the Bayesian KMO (BKMO) index from a posterior of correlations estimated with a Bayesian GLM (Section 3.2). To compare the BKMO index to its

frequentist counterpart, a series of simulations are conducted that systematically varies important characteristics of the data matrix (Section 3.3).

3.1 Reconceptualization

To reconceptualize the KMO index in a 'proper' Bayesian manner, it is necessary to consider *Bayes' theorem* (Bayes and Price, 1763; Laplace, 1814), sometimes known as *Bayes' rule* (e.g., Kruschke, 2014: 100-104), which is the conceptual foundation of Bayesian inference (Gelman et al., 2014). As shown in equation (3.1.1) in relation to two discrete events (a and b), Bayes' theorem consists of $P(a)$, which is the *prior probability*; $P(b)$, the *marginal likelihood*; $P(b | a)$, the *likelihood*; and $P(a | b)$, the *posterior probability* (McElreath, 2019: 49).

$$P(a | b) = \frac{P(a)P(b | a)}{P(b)} \quad (3.1.1)$$

Consistent with Bayesian inference (e.g., Levy and Mislevy, 2020), a and b in equation (3.1.1) will here be respectively reconceptualized as the 'sampling adequacy' (i.e., α_{Bayes}) and data matrix ($\mathcal{D}$). Here α_{Bayes} substitutes the frequentist concept of the 'sampling adequacy' *in the population* (α) considered in section 2.1, since this Bayesian conceptualization instead will reflect the 'sampling adequacy' *given the data*, where α may *not* equal α_{Bayes} due to differing formula (e.g., the presence of prior information). This reconceptualization of a and b in Bayes' theorem is appropriate, since the formula in section 2.1 showed that the 'sampling adequacy' is a function of the inter-correlations ($\hat{\rho}$) and anti-image inter-correlations ($\hat{\rho}_q$), which both are based on the data matrix ($\mathcal{D}$). For similar reasons, the inter-correlations, inverse inter-correlations, and anti-image inter-correlations, when infused with prior information, can be distinctly denoted $\hat{\rho}_{\text{Bayes}}$, $\hat{\rho}_{\text{Bayes}}^{-1}$, and $\hat{\rho}_{q\text{Bayes}}$, respectively.

In equation (3.1.1), $P(a)$ can then be denoted $P(\alpha_{\text{Bayes}})$ and conceptualized as the *prior probability of the 'sampling adequacy'*, which can be more meaningfully interpreted as the uncertainty about the factorability of $\mathcal{D}$ before observing it. Such prior uncertainty can be taken to reflect the stochasticity involved in data collection (e.g., random sampling can produce data of varying quality) and/or fundamental epistemological uncertainty about the phenomenon being studied.

$P(b)$ can then be denoted $P(\mathcal{D})$ and becomes the *marginal likelihood of the data matrix*. While this factor can be difficult, if not practically impossible, to calculate, except for only the simplest of models (cf. Kruschke, 2014: 115), recent advances in *Markov-Chain Monte Carlo* methods (MCMC; Brooks et al., 2011) have solved this limitation by exploiting the fact that $P(a | b) \propto P(a)P(b | a)$, since $P(b)$ is merely a normalizing constant, leading to an estimation procedure that entirely circumvents its calculation. Computational difficulties that could arise from a Bayesian version of the KMO index is thus here mitigated by building the reconceptualization around a reliance on these easily applicable MCMC methods.

Accordingly, $P(b | a)$ becomes $P(\mathcal{D} | \alpha_{\text{Bayes}})$, which reflects the *likelihood of the data matrix given the 'sampling adequacy'*, and from this, it follows that $P(\alpha_{\text{Bayes}} | \mathcal{D})$ can be conceptualized as the *posterior probability of the 'sampling adequacy'*, that is, the probability of the 'sampling adequacy' given the data.

With the reliance on MCMC methods, researchers thus only need to specify the prior of the sampling adequacy, $P(\alpha_{\text{Bayes}})$, and the likelihood of the data given the 'sampling adequacy', $P(\mathcal{D} | \alpha_{\text{Bayes}})$, to arrive at the posterior probability of the 'sampling adequacy', $P(\alpha_{\text{Bayes}} | \mathcal{D})$. However, considering the relatively abstract concept and interpretation of 'sampling adequacy', researchers may *not* find it easy to specify a prior and likelihood distribution

for it.

These difficulties can be circumvented by recognizing that researchers need *not* necessarily specify a prior and likelihood for the KMO index *per se*, which is contrary to the specification of most statistical models in Bayesian inference (e.g., McElreath, 2019; for an exception, see, e.g., Kruschke, 2014: 226-230). This circumvention is possible because the KMO index can be recognized as *not* just a monotonous function of the correlation matrix, but a *deterministic* function of it, meaning that there is no *additional* stochastic nor epistemological uncertainty in transitioning from a prior on $\hat{P}$ – through $\hat{P}^{-1}$ and $\hat{P}_q$ – to a prior on $\hat{\alpha}$ and $\hat{\alpha}_m$. Stated differently, the prior and likelihood of the correlation matrix *implies* a prior and likelihood of the KMO index, and a prior and likelihood for solely the former are therefore sufficient. Estimating the Bayesian KMO index thus becomes a question of applying Bayesian inference in relation to a correlation matrix.

Since a correlation matrix is inherently a standardized covariance matrix containing the bivariate relationships between the k items of interest, Bayesian inference can be applied by simply rescaling the data using *Fisher z-scoring* (henceforth *standardization*; Field, 2018: 37), and then, for each of the $\frac{k(k-1)}{2}$ bivariate relationships, computing the *standardized coefficient*, which is equivalent to the correlation coefficient due to the standardization (cf. Agresti, 2018: 273). These standardized coefficients can then be organized to produce a correlation matrix.

Such an approach can be efficiently modeled using a *generalized linear model* (GLM; Field, 2018; Kruschke, 2014), which require the specification of a likelihood function. As the product-moment correlation coefficient is a measure of association that assumes a linear-normal relationship between variables, an appropriate likelihood for this application of the GLM is the *multivariate Gaussian* (i.e., normal) *probability distribution* (Gelman et al., 2014: 578, 582). By using this distribution as the likelihood, it is apparent that the resulting Bayesian GLM (BGLM) will invoke standard assumptions, such as linearity in the parameters, no missing data, zero conditional mean, homoskedasticity, and the absence of outliers (cf. Stock and Watson, 2019; Wooldridge, 2019).⁹

With the multivariate normal in mind for the likelihood, this leaves the question of specifying the priors for the correlation matrix. A natural choice for such a prior is the *Lewandowski-Kurowicka-Joe* probability distribution ($\mathcal{LKJ}$; Lewandowski et al., 2009), which is an appropriate and often used prior for correlation matrices in Bayesian inference (e.g., McElreath, 2019: 441-442). The priors can be specified with this $\mathcal{LKJ}$ distribution through use of its single *shape* parameter (i.e., η), which is a positive real scalar. When $\eta = 1$, the distribution of correlations is flat; with $\eta > 1$, the distribution of correlations is less extreme, approaching an *identity matrix* (Boyd and Vandenberghe, 2018: 113) at greater values; and with $\eta < 1$, the distribution becomes more extreme, approaching a singular correlation matrix at lower values (McElreath, 2019: 442). This means that the $\mathcal{LKJ}$ probability distribution can be used to specify a joint prior for all the correlations, making it easy for researchers to specify priors with many measured items.

To reflect the expected lack of substantial prior information about the inter-correlations that characterizes an *exploratory factor analysis* (EFA; Field, 2018: 778-833), researchers should ideally specify the shape parameter to produce a 'weak' or 'non-informative' prior (cf. Gelman et al., 2014: 51-56). In line with Gelman and colleagues (2014: 51-52), the author recommends specifying 'weakly-informative' priors, which can be done with the $\mathcal{LKJ}$ probability distribution in numerous ways. One general approach involves specifying 'skeptical, yet persuadable'

⁹For researchers interested in greater robustness against violations of assumptions, a more robust BKMO index is discussed and derived in section 8.3 of the appendix.

priors, whose distributions are deliberately conservative by expecting small inter-correlations but which remain wide enough to 'let data speak for itself' (e.g., $\eta = 2$; McElreath, 2019: 441-442). Such priors can impact data in a manner similar to *regularization* by reducing *overfitting*, which can improve the generalization of results (cf. Kruschke, 2014: 531-532; McElreath, 2019: 214-217). However, in the event that researchers do possess quality prior information, they should *not* stray from using this information to specify 'strongly informative' priors (cf. Kruschke, 2014: 113-115), though they should always transparently report their priors to open them up to criticism from their peers.

Regardless of prior specification, researchers may be concerned whether their choice of priors inadvertently influences the results. As a response to this, it can be stated that for most statistical models, as long as the prior assigns non-zero probability to every possible parameter value, that the data can quickly overwhelm the prior (cf. Kruschke, 2014: 293; Wagenmakers et al., 2010: 167), producing relatively prior-insensitive results that closely resemble frequentist and likelihood-based estimates but for whom a Bayesian interpretation can be made. Similarly, should researchers specify priors using a uniform distribution (e.g., $\eta = 1$, McElreath, 2019: 441-442), results will typically approximate likelihood-based estimates (cf. Hastie et al., 2017: 272; King, 1998: 30) even with little data (cf. Kruschke, 2014: 113). Researchers can also compare results across different priors and/or to frequentist results derived from the *non-parametric bootstrap* (Efron, 1979, 2003; Efron and Tibshirani, 1994), to assess whether results are robust to the prior specification and statistical framework, though researchers should note that the re-sampling involved in the non-parametric bootstrap is *not* equivalent to the MCMC method of Bayesian inference (cf. Kruschke, 2014: 161).

With these considerations for the specification of the prior in mind, the *Bayesian KMO* (BKMO) *index* is thus conceptualized as 'a measure of sampling adequacy *given* the data', which may be denoted α_{Bayes} , or alternatively α , depending on the need to distinguish it from its frequentist conceptualization. As mentioned, however, since this concept is *not* analytically derived, but instead estimated with MCMC methods, this *estimated* 'sampling adequacy' should be appropriately distinguished, for example, as $\hat{\alpha}_{\text{Bayes}}$ (or $\hat{\alpha}$).

3.2 Formula

By now, it should have been made clear that the BKMO index is only indirectly computed using MCMC methods, and instead is to be calculated directly from the posterior distribution of correlation matrices resulting from a Bayesian GLM (BGLM), which is why the likelihood and prior solely needed to be specified for the correlation matrix. This means that the BKMO index can be calculated by simply, for each MCMC-draw, constructing a correlation matrix from the inter-correlations estimated by the BGLM, deriving the anti-image correlation matrix from the inverse correlation matrix, followed by calculating the overall and individual KMO indices.

Accordingly, the formula for the KMO index reviewed in section 2.1 can be adjusted for this Bayesian reconceptualization, which is appropriate, because all prior information has already been incorporated into the estimated posterior correlation matrix (i.e., $\hat{P}_{\text{Bayes}}$). Concisely expressed, calculating the overall BKMO index is achieved with function (3.2.1):

$$\text{BKMO}(\mathcal{D}) \equiv \hat{\alpha}_{\text{Bayes}[d]} \equiv \frac{\sum_{i \neq j} \sum \hat{\rho}_{\text{Bayes}[i,j,d]}^2}{\sum_{i \neq j} \sum \hat{\rho}_{\text{Bayes}[i,j,d]}^2 + \sum_{i \neq j} \sum \hat{\rho}_{q\text{Bayes}[i,j,d]}^2} \quad (3.2.1)$$

where $\hat{\alpha}_{\text{Bayes}}$ is the estimated overall 'sampling adequacy', d denotes the index for the d th posterior draw, $\hat{\rho}_{\text{Bayes}}$ is the estimated correlation coefficient, while $\hat{\rho}_{q\text{Bayes}}$ is the estimated anti-image correlation coefficient.

Similarly, calculating the individual BKMO indices can be accomplished with function (3.2.2):

$$\text{BKMO}(\mathcal{D}_m) \equiv \hat{\alpha}_{\text{Bayes}[m,d]} \equiv \frac{\sum_{j,j \neq i} \hat{\rho}_{\text{Bayes}[i,j,d]}^2}{\sum_{j,j \neq i} \hat{\rho}_{\text{Bayes}[i,j,d]}^2 + \sum_{j,j \neq i} \hat{\rho}_{q\text{Bayes}[i,j,d]}^2} \quad (3.2.2)$$

where $\hat{\alpha}_{\text{Bayes}[m]}$ is the estimated 'sampling adequacy' for item m .

In accordance with this Bayesian reconceptualization of the KMO index, from the distribution of its estimated posterior values, coherent probabilistic statements can be made about the 'sampling adequacy' given the data. For example, one can ascertain the posterior probability that the 'sampling adequacy' is greater than .5 given the data, $P(\hat{\alpha}_{\text{Bayes}} > .5 \mid \mathcal{D})$, or one can compute the 95% *highest density (contiguous) interval* (HD(C)I; Kay, 2024a; 2024b; Makowski et al., 2019b), which contains the 95% most probable values of the 'sampling adequacy' given the data, both of which are coherent probabilistic statements within the Bayesian framework (cf. Kruschke, 2014). This is conceptually different from the frequentist KMO index, which instead provides inferential statements about the data given the 'sampling adequacy', $P(\mathcal{D} \mid \alpha)$.

3.3 Sampling behavior

To assess the sampling behavior of the Bayesian KMO (BKMO) index and compare it to its frequentist counterpart, a series of simulations were set up that systematically varied the sample size, priors, magnitude of the factor loadings, number of factors, and the number of items. This section only briefly details the implementation of the BKMO index and focuses on the results from these systematic comparisons, with the following section instead providing greater detail on how to practically implement the BKMO index with empirical data (Section 4).

All the simulations conducted involved specifying a standard normally-distributed factor at the level of a population, which influenced a number of items measured in a data matrix. This approach meant that the 'true' factorability was known in advance and that the frequentist and Bayesian KMO indices could be computed using repeated samples of varying size and compared to the known sampling adequacy using the *root mean squared error* (RMSE). For realism, the population size was specified to be 1,000,000 units, while the sample sizes ranged from 30 to 1,000, with 30 chosen as the lower limit to mitigate issues with multicollinearity at small samples, while 1,000 constitutes a sample size where one could expect a reasonable prior to be dominated by the data.

To assess the impact of different $\mathcal{LKJ}$ -prior specifications, multiple BGLMs were fit across all simulations, each using either a 'liberal' (i.e., $\eta = 0.5$), 'noninformative' (i.e., $\eta = 1$), or 'conservative' (i.e., $\eta = 2$) prior. Given the computational costs of fitting Bayesian models, only 100 simulations were conducted across the range of sample sizes, and the number of posterior draws per chain were limited to 1,000, which has previously proven adequate for similar assessment (see, e.g., Vehtari et al., 2016: 1419). Since repeated inspections with a relatively large number of independent sampling chains (i.e., 15) found results consistent with convergence after merely 100 warm-up draws, computation was made faster by only using 1 sampling chain and 100 warm-up draws. To assess variability of the estimates, the frequentist KMO indices exploited 1,000 non-parametric bootstrapped samples, while the BKMO indices used the 1,000 posterior draws.¹⁰

¹⁰For more information about the simulations, see the appendix (Section 8).

The first series of simulations involved a condition where the number of factors (i.e., 1), items (i.e., 3), and magnitude of the relationship between the factor and items (i.e., $\beta = 0.487$) were fixed (with was specified to provide a 80% statistical power at the 5% level for the smallest sample size), while solely the sample size was allowed to vary. Whether frequentist or Bayesian, the overall and individual KMO indices all showed signs of consistency, with the RMSE appearing to approach 0 as the sample size increased. Irrespective of prior specification, all differences between the KMO and BKMO indices were *statistically indiscernible* from each other at the 95% certainty level, even at the smallest sample size (i.e., $\Delta\hat{\alpha} = .004$, $SE = .084$, 95% CI[-.159; .177]), corroborating the aforementioned notion that Bayesian inference can approximate frequentist results in small samples.

The second series of simulations used the same specifications, except that the factor-item relationship for all items was first fixed to 0 in one condition and then to twice the coefficient from the first condition (i.e., $\beta = 0.973$). Similar sampling behavior was observed, though the BKMO index, irrespective of prior, showed signs of less bias compared to the KMO index at smaller sample sizes when the coefficient was 0. The inverse held for a coefficient of 0.973, where the KMO index appeared less biased at smaller sample sizes. While this could be considered surprising, none of these differences proved statistically discernible, even where the difference was largest (i.e., $\Delta\hat{\alpha} = -.011$, $SE = .059$, 95% CI[-.132; .102]). Compared to the first condition, where the coefficient had been fixed to 0.487, there were otherwise no signs of any differences to the sampling characteristics already observed.

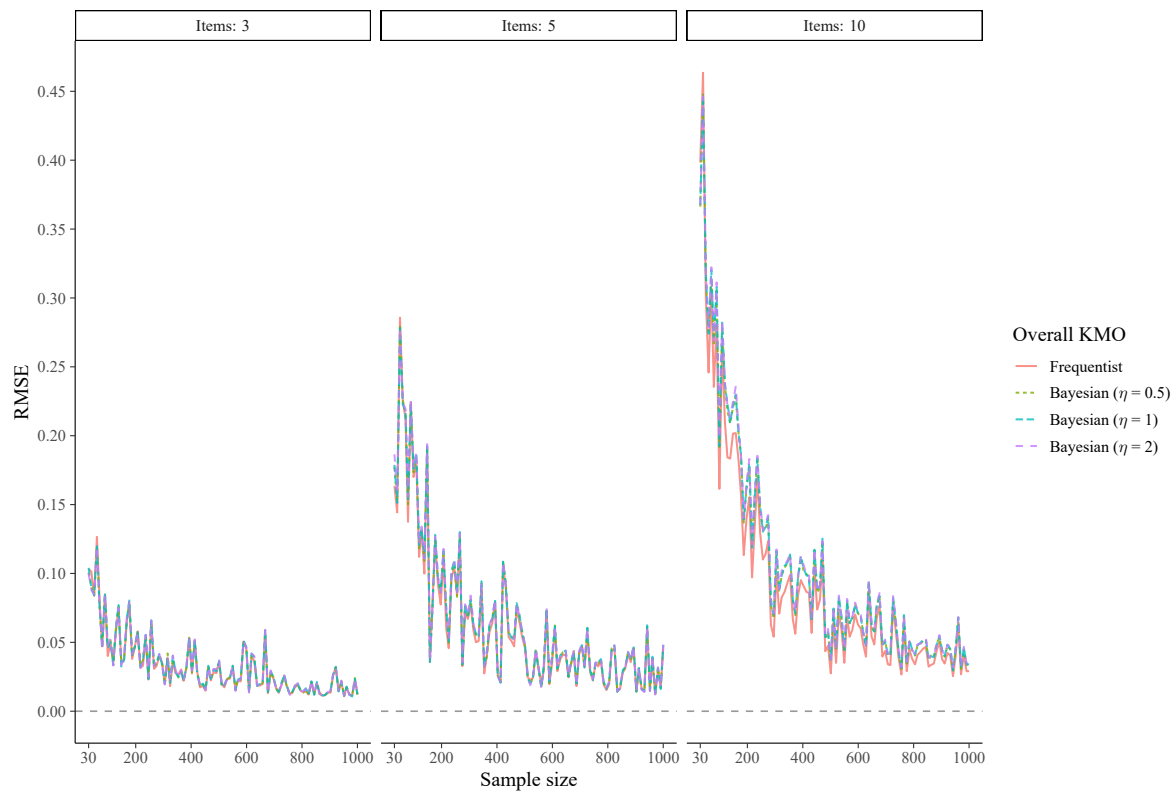


Figure 1. Sampling characteristics of the KMO and BKMO index

NOTE: The root mean squared error (RMSE) of the overall KMO and BKMO indices across varying sample sizes (i.e., 30 – 1000), number of items (i.e., 3 – 10), and η -parameter values for the $\mathcal{LKJ}$ -prior (i.e., 0.5 – 2). Number of latent factors = 1, factor-item coefficient = 0.487. Simulations = 100. Posterior/Bootstrap samples = 1,000.

The third series of simulations were similar to the first, with the exception of the number of items being increased first from 3 to 5 in one condition and then to 10 in another. The results of these simulations are provided

in figure 1. While the KMO index appeared to have a lower RMSE at lower sample sizes when the number of items increased, none of its differences with the BKMO indices, irrespective of prior, were statistically discernible, including at the sample size (i.e., $n = 128$) where the difference appeared largest (i.e., $\Delta\hat{\alpha} = .041$, $SE = .071$, 95% $CI[-.107; .177]$). Compared to the first condition with only 3 items, the increase in items seemed to increase the RMSE for small samples, though it had little to no effect on larger samples. Similar sampling behavior was otherwise observed, consistent with the notion that the sampling behavior of the KMO and BKMO indices are highly similar.

The fourth and last series of simulations instead involved increasing the number of factors, first to 2 in one condition (i.e., one factor influences 2 items, one factor influences 1) and then to 3 in another (i.e., each factor influences one item). While the RMSE of the KMO index appeared to increase as the number of factors grew, none of the differences were statistically discernible, even when the difference appeared to be largest (i.e., $\Delta\hat{\alpha} = -.011$, $SE = .059$, 95% $CI[-.132; .102]$). No other differences to the already established sampling behaviors were otherwise identified.

By systematically varying the characteristics of the data matrix, these simulations have corroborated the aforementioned notion that the KMO and BKMO indices provide similar estimates of factorability, even in small samples, and that the results are relatively insensitive to the prior specification, with even relatively little data overwhelming the specified priors. These results are therefore consistent with the notion that the KMO and BKMO indices mainly differ conceptually, though researchers will generally be able to compute both and compare them within their specific research context.

4 Demonstration

Having revisited the classical KMO index and conceptualized a Bayesian KMO index in the sections thus far, this section serves to demonstrate how researchers can use this BKMO index on empirical data to assess the 'sampling adequacy' given that data. This involves characterizing the empirical data from which the BKMO index is to be computed (Section 4.1); applying a BGLM with 'weakly informative' priors to estimate the correlations from which the BKMO index is computed (Section 4.2); an assessment of the sensitivity of results to the prior specification (Section 4.3); and an empirical comparison between the frequentist KMO index and the BKMO index (Section 4.4). All data processing and analyses in this section were achieved with the R programming language (R Core Team, 2025), using the RStudio IDE (Posit Team, 2025).¹¹

4.1 Empirical Data

For the purposes of this demonstration, empirical data by Rosenberg (1965; provided in O'Connor, 2024) is used. This data consists of a 300 by 10 matrix, where each row is a series of measured responses by a volunteer high school student from New York, while each column is their response on a 4-point *Likert* scale (Likert, 1932) towards an item from the validated *Rosenberg self-esteem scale* (e.g., 'On the whole, I am satisfied with myself'; Rosenberg, 1965: 17), which serves to measure the theorized unidimensional latent phenomenon of *self-esteem*.

That data is used here for several reasons. Besides being popular for demonstrative purposes (e.g., O'Connor, 2024), it has no missing data, while the use of 10 items to measure the manifestations of self-esteem is consistent

¹¹For software specifications and replication materials, see the appendix (Section 8).

with the number of items in other successfully validated psychometric inventories (e.g., Ashton et al., 2007), and the 300 observations in the data also provides a *statistical power* (Cohen, 1988, 1991) of approximately 99.9% to detect a standardized factor loading as low as .3, which can be considered the *smallest effect size of interest* (SESOI; Lakens, 2017) in relation to factor analysis (cf. Field, 2018: 806).

4.2 BKMO index

Following the BGLM-modeling approach outlined in section 3.1, all items are first standardized. A common Bayesian expression of a GLM (Levy and Mislevy, 2020; McElreath, 2019: 441-442), which is used to compute the inter-correlations between the items in the data is then formalized as model (3.3.1).

$$Z(\theta)_u | \mu_m, \Omega \sim \mathcal{MVN}(\mu_m, \Omega) \text{ for } u \text{ in } 1, \dots, n \quad (3.3.1)$$

$$\Omega = DPD$$

$$\mu_m = 0 \text{ for } m \text{ in } 1, \dots, k$$

$$\text{diag}(D)_m = 1 \text{ for } m \text{ in } 1, \dots, k$$

$$P \sim \mathcal{LKJ}(2)$$

This statistical model states that the individual standardized items (i.e., $Z(\theta)_u$) in the data matrix ($\mathcal{D}$) are *iid*¹² as a conditionally multivariate normal probability distribution ($\mathcal{MVN}$), with a k -length item-varying mean vector parameter (i.e., μ_m) and a k by k variance-covariance matrix (i.e., Ω) parameter.

Following Barnard and colleagues (2000; see also McElreath, 2019: 441-442), the variance-covariance matrix of model (3.3.1) is decomposed into two k by k *diagonal matrices* (Boyd and Vandenberghe, 2018: 114) of standard deviations (i.e., D) and one k by k correlation matrix (i.e., P). Due to the aforementioned standardization of the data, and the absence of predictors in the model, the mean vector becomes a k -length zero vector of intercepts. D similarly becomes a k by k identity matrix, because its diagonal becomes a *ones vector* (cf. Boyd and Vandenberghe, 2018: 5), which simplifies the variance-covariance matrix (Ω) into the correlation matrix (i.e., P). As such, due to the standardization, squareness, symmetry, and ones vector diagonal that characterizes a correlation matrix, this modeling approach is useful, because the researcher is technically only required to formulate priors for, and estimate, the unique $\frac{k(k-1)}{2}$ inter-correlations in P , which can speed up computation.

The prior distribution of P in model (3.3.1) is specified using the $\mathcal{LKJ}$ probability distribution, whose shape parameter (η) is specified as 2 to reflect an *a priori* belief in relatively small inter-correlations, which is intended to be 'regularizing/weakly informative' relative to the data (cf. McElreath, 2019: 442).

To visualize the prior for the 'sampling adequacy' implied by model (3.3.1), MCMC can be used to sample from its prior distribution (Kruschke, 2014: 212). This is done with the *brms* R package (Bürkner, 2017, 2018), which is useful due to its flexibility in model specification and reliance on the Stan probabilistic programming language (Carpenter et al., 2017; Stan Development Team, 2025a; 2025b) that enables efficient sampling from the posterior using the *no-u-turn sampler* (Hoffman and Gelman, 2014) for Hamiltonian MCMC (cf. Kruschke, 2014: 400-416). The MCMC specifications are further detailed below.

¹²Consistent with the Bayesian framework (Levy and Mislevy, 2020), the relatively stringent *iid* assumption in model (3.3.1) may be relaxed by replacing it with the weaker assumption of *exchangeability* (de Finetti, 1974). This means that the units could be interdependent (e.g., by belonging to distinct clusters), but that the priors and model treat them as exchangeable given the available information (cf. Levy and Mislevy, 2020: 56-64; McElreath, 2019: 81).

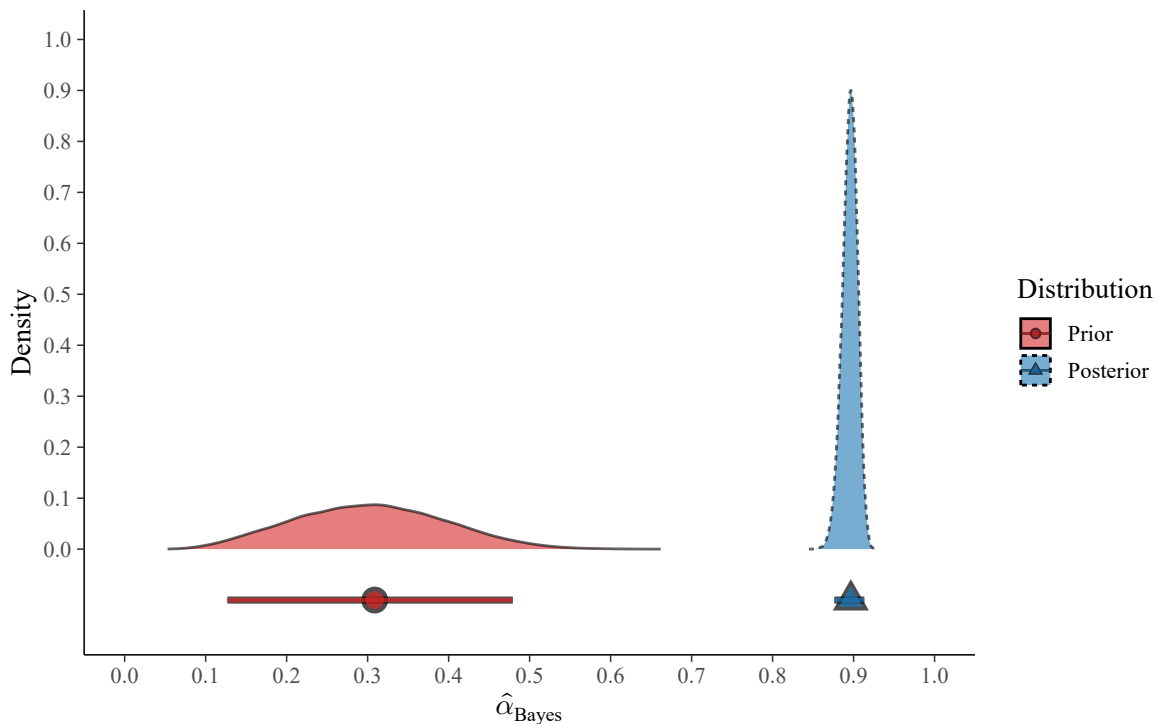


Figure 2. Overall ‘sampling adequacy’

NOTE: Prior/posterior distributions of the estimated overall Bayesian KMO index (i.e., $\hat{\alpha}_{\text{Bayes}}$). Prior/posterior samples = 40,005. The geometrics below the distributions indicate the mode (circle) and the 95% HDI (bar). Results derived from data by Rosenberg (1965; provided in O’Connor, 2024).

The implied prior distribution of the overall ‘sampling adequacy’ resulting from this $\mathcal{LKJ}$ prior has an average of .304 (SD = .093; 95% HDI[.127; .480]), with the estimated distribution being shown in figure 2. Inspecting this prior reveals an implied *a priori* belief in a relatively low ‘sampling adequacy’. This can be substantively interpreted to reflect a conservative expectation that the data will be relatively *infactorable* due to a high diffusion in correlations and hence be inappropriate for a subsequent factor analysis. In this instance, a shape parameter for the $\mathcal{LKJ}$ prior distribution of 2 for the 10 theorized manifestations thus implies a ‘skeptical’ prior for the ‘sampling adequacy’.

To estimate the posterior, model (3.3.1) is then fit to the data using the same software. To help assess whether the sampler converged to the posterior distribution, 15 independent sampling chains were used, and 2,000 warm-up samples were drawn to increase the chances of convergence prior to sampling. After the warm-up period, 2,667 samples were drawn from each chain, resulting in a total of 40,005 posterior samples, whose size was chosen to help invoke the *law of large numbers* (Stock and Watson, 2019: 85-86), as well as ensure a sufficient *effective sample size* when accounting for *autocorrelation* (i.e., ESS $\geq$ 10,000; Kruschke, 2014: 182-184). Based on visual inspections of *trace plots* (Kruschke, 2014: 179-180), adequate values of the *potential scale reduction factor* ($\hat{R} < 1.01$; Vehtari et al., 2021) and the *Monte Carlo Standard Error* (i.e., MCSE $\leq$ 0.001; Kruschke, 2014: 186-187), the sampling chains were consistent with convergence and thus the posterior distribution seemed to have been reliably identified.

Having estimated the posterior distribution of the inter-correlations, the BKMO index can then be computed using the procedure outlined in section 3.2. Since the posterior distribution consists of 40,005 samples, the function implementing the calculation of the BKMO relied on *vectorization* to speed up computation (cf. Kruschke, 2014:

408). This BKMO index function was implemented using R code and is provided for ease of implementation in section 8.2 of the appendix.

The estimated posterior distribution of the BKMO index, that is, the estimated 'sampling adequacy' given the data matrix ($\hat{\alpha}_{\text{Bayes}}$), is provided visually next to the prior distribution in figure 2. These results can be summarized metrically, with the posterior average of the overall BKMO index being .895, with a posterior standard deviation¹³ (SD) of .010. Computing the 95% HDI reveals that the 'sampling adequacy' with 95% credibility is between .876 and .913 given the data. This means that the diffusion in the correlations are relatively low with a high degree of credibility, and the data matrix can be considered factorable and appropriate for a subsequent factor analysis.

Comparing the posterior to the prior distribution reveals an overlap of approximately .000%. With an average difference of .592 (SD = .093; 95% HDI[.417; .771]), the two distributions can be considered credibly different from each other, indicating that the prior information about the 'sampling adequacy' has been credibly updated by this data.

4.3 Prior sensitivity

To demonstrate how researchers can assess the sensitivity of results to their choice of priors, the posterior derived from 'strongly' or 'weakly informative' priors can be compared to results derived from 'non-informative' priors.

This can be demonstrated here by respecifying model (3.3.1) so that the $\mathcal{LKJ}$ prior has a shape parameter of 1, creating a flat prior probability for the correlation matrix, which as previously mentioned, makes it approximate a maximum-likelihood model. Refitting it yields reliable model diagnostics similar to the two aforementioned models, which are *not* reported here for brevity. Its distribution is illustrated in figure 3, where the average estimated overall BKMO index value is .896 (SD = .010; 95% HDI[.877; .914]).

The overlap between this posterior distribution with a 'non-informative' prior and the previous posterior with a 'weakly informative' prior is approximately 97%, and they have an average difference of -.001 (SD = .013; 95% HDI[-.028; .025]), revealing them to be credibly *indifferent*. This indicates that the prior specified in model (3.3.1) is 'persuadable'/'weakly informative' relative to this data, and it further corroborates that the BKMO index can approximate likelihood-based results with varying priors and enough data.

4.4 Frequentist comparison

The BKMO index can also be compared to the frequentist KMO index to demonstrate that conclusions are robust across frameworks. A meaningful comparison can be achieved by applying a non-parametric bootstrap to the computation of the frequentist KMO index to derive an estimate of its sampling distribution, which similarly approximates MCMC with 'non-informative' priors.

To achieve this, the data can be permuted a sufficiently large number of times (i.e., the rows of $\mathcal{D}$ are re-sampled with replacement) to produce a distribution of counterfactually sampled data matrices. From each permuted data matrix, an overall and k individual KMO indices can be calculated by applying the functions in section 2.1, to produce an approximation of the sampling distribution of the KMO indices. Here, the mean, *bias-reduced standard error* (SE; cf. Park et al., 2022; Park and Wang, 2020, 2022), and 95% *bias-corrected and accelerated interval* (BCa; Efron, 1987) are calculated from 40,005 re-samples of the data. This number of permutations was

¹³Consistent the Bayesian approach (McElreath, 2019: 197), the standard deviation of the posterior distribution is derived without applying Bessel's correction when computing the variance (i.e., $\frac{1}{n-1}$ is substituted with $\frac{1}{n}$).

chosen to help invoke the law of large numbers and for comparability with results from the BKMO index.

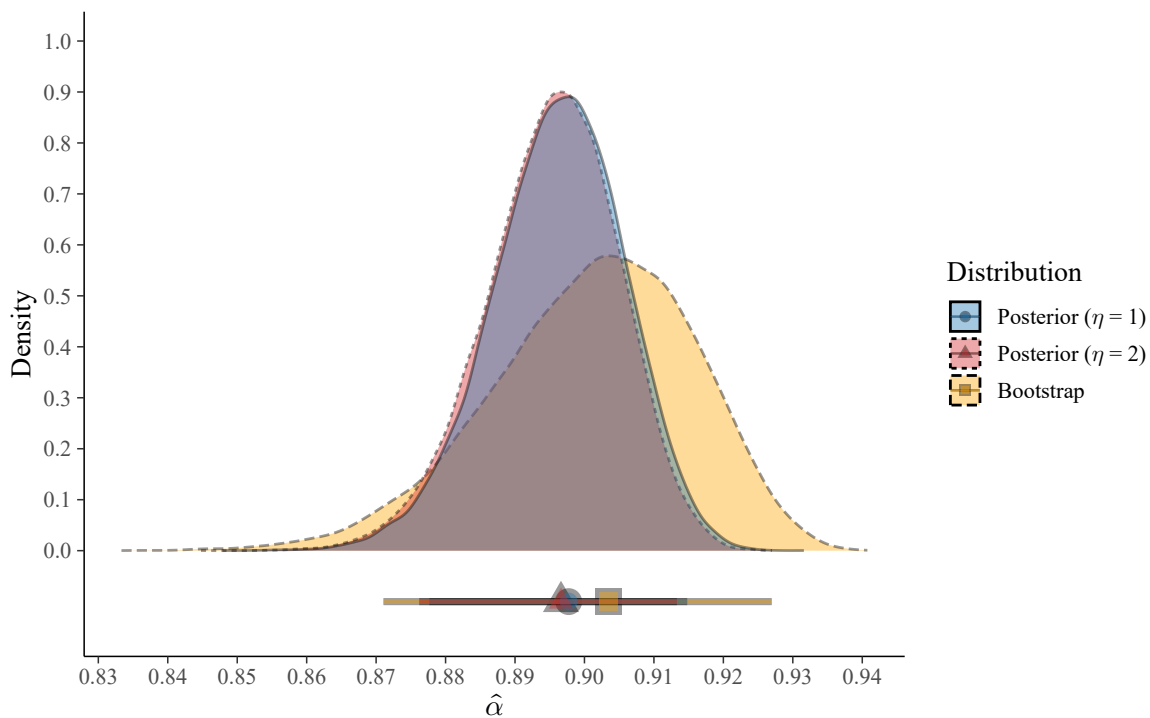


Figure 3. Comparison of overall 'sampling adequacy' across distributions

NOTE: Posterior with $\eta = 2$, posterior with $\eta = 1$, and (non-parametric) bootstrap distributions of the estimated overall KMO index (i.e., $\hat{\alpha}$). Posterior/bootstrap samples = 40,005. The geometrics below the distributions indicate the mode (circle) and the 95% HDI (bar). Results derived from data by Rosenberg (1965; provided in O'Connor, 2024).

The distribution of results from this non-parametric bootstrap-based approach is provided in figure 3 alongside the previously mentioned posterior distributions. This bootstrap distribution has an average estimated overall KMO index of .901 (SE = .015; 95% BCa[.866; .924]), where the SE and 95% BCa, given their frequentist conceptualization, help express the (sampling) uncertainty about the estimated 'sampling adequacy' in the population.

The overlap between this bootstrap distribution and the posterior with a 'weakly informative' prior (i.e., $\eta = 2$) is approximately 70.5%, with an average difference of -.006 (SD = .018; 95% HDI[-.039; .029]). This relatively high degree of overlap and lack of a credible difference is again consistent with results being insensitive to a frequentist or Bayesian approach, and it shows that the prior used for the posterior distribution is 'weakly informative', further corroborating that the BKMO index can approximate a bootstrapped frequentist KMO index.

As such, a practical application of the BKMO index has been demonstrated in relation to empirical data, making its ease of use evident. Comparisons between different priors and the frequentist KMO index further corroborated that conclusions about the 'sampling adequacy' of the data are relatively insensitive to the prior specification and statistical framework. Results from the BKMO index can thus approximate the frequentist KMO index, though solely the Bayesian version enables the incorporation of prior information and the ability to form coherent probabilistic statements from the posterior about the factorability of the data.

5 Discussion

The conceptualized Bayesian KMO (BKMO) index introduced in this paper can be considered a novel contribution to the literature on psychometrics, particularly the subdiscipline of factor analysis. While a KMO index already exists that can provide critical information to researchers developing new psychometric inventories, this classical KMO index was demonstrated to be frequentist, which makes it conceptually incompatible with Bayesian inference. This is made evident by the conceptualized BKMO index being uniquely able to incorporate prior information when assessing the 'sampling adequacy' of a data matrix, and from its posterior distribution, researchers can make coherent probabilistic statements about the factorability of the data matrix given that data.

Because the BKMO index is a Bayesian concept, statistical inference in relation to it requires priors, which can pose a point of contention between researchers. As discussed and demonstrated, however, this issue is generally mitigated even with a relatively small sample size, and researchers can demonstrate that conclusions are robust by comparing results derived from different priors. A point of contention may also arise from the choice of using frequentist or Bayesian inference, but researchers can similarly demonstrate that conclusions are robust by comparing results derived from a frequentist KMO index to the BKMO index. Irrespective of priors and statistical framework, researchers should always be wary of violations of the assumptions underlying both the KMO and BKMO indices (e.g., multicollinearity), since these could prevent a meaningful assessment of the factorability of the data matrix.

Like its frequentist counterpart, the areas of application for the BKMO index are numerous, and to remain consistent with their statistical framework, Bayesian researchers should ideally substitute the frequentist KMO index for its Bayesian counterpart whenever they are interested in assessing the 'sampling adequacy' of a data matrix prior to a factor analysis. While the 'sampling adequacy' is generally assessed prior to an EFA (Field, 2018: 778-833), the BKMO index could be used in relation to a *confirmatory factor analysis* (CFA; Brown, 2015; Levy and Mislevy, 2020), where researchers, by merit of possessing more information about the theorized factor structure, could better justify specifying stronger priors. Here, it could be meaningful to also prespecify and test hypotheses about the 'sampling adequacy' of the data matrix prior to a CFA, and the BKMO index would enable researchers to use such prior information to better test these hypotheses, something that researchers will be incapable of with the classical KMO index.

6 Conclusion

This paper has introduced the Bayesian KMO index. This was accomplished by reviewing the classical Kaiser-Meyer-Olkin (KMO) index, which is a measure of sampling adequacy (MSA) that helps assess whether a data matrix is appropriate for a factor analysis. To build towards a Bayesian KMO index, this review revisited its calculation and discussed its mathematical and conceptual properties, which served to show that the KMO index is a frequentist statistic. Bayes' theorem was then used to reconceptualize the KMO index, resulting in a Bayesian KMO (BKMO) index that can incorporate prior information and enable coherent probabilistic statements to be made about the 'sampling adequacy' given the data. Systematic comparisons using simulated data corroborated the notion that the BKMO index approximates its frequentist counterpart, with results generally being insensitive to different prior specifications, even at relatively small sample sizes. Estimated with MCMC methods, its ease of

computation was demonstrated on empirical data. To enable researchers to implement this BKMO index on their own, this paper provides efficient code to easily implement it in R software.

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8 Appendix

This appendix contains the specifications for software used in this paper (Section 8.1), the R function implementing the Bayesian Kaiser-Meyer-Olkin (BKMO) index (Section 8.2) and a brief discussion on deriving a robust BKMO index (Section 8.3). Besides this appendix, supplementary materials, including software specifications, replication files, and the simulated data used in section 3.3 are made freely available online on the OSF project page dedicated to this paper (DOI: 10.17605/OSF.IO/T3UPD).

8.1 Software specifications

This document was written with *markdown* (Gruber, 2014) and L^AT_EX (Lamport, 1986; The LaTeX Project, 2025), using Xe_TE_X(v3.141592653-2.6-0.999998; Kew and Hosny, 2023), pdf_TE_X(v3.141592653-2.6-1.40.29; et al., 2026) and the TeXmaker IDE (v6.0.1; Brachet, 2025). Compiled on a Windows 11 x64 (build 26200) operating system (OS; Microsoft, 2024) on the 2026-04-04 10:09 CEST (Timezone: Europe/Copenhagen), the machine has 16 cores available, and the range of doubles is 2.23×10^{-308} - $1.8 \times 10^{+308}$.

All data processing and analyses were conducted on the aforementioned software using the R programming language (v4.4.3 [2025-02-28 ucrt]; R Core Team, 2025) in the RStudio IDE (2025.5.1.513; Posit Team, 2025), using the R packages bayesplot (v1.13.0; Gabry et al., 2019), bayestestR (v0.16.1; Makowski et al., 2019b), bookdown (v0.43; Xie, 2016, 2025a), brms (v2.22.0; Bürkner, 2017, 2018), datawizard (v1.1.0; Patil et al., 2022), doParallel (v1.0.17; Microsoft and Weston, 2022a), dplyr (v1.1.4; Wickham et al., 2023), EFA.dimensions (v0.1.8.4; O'Connor, 2024), effectsize (v1.0.1; Ben-Shachar et al., 2020), foreach (v1.5.2; Microsoft and Weston, 2022b), ggplot2 (v3.5.2; Wickham, 2016), kableExtra (v1.4.0; Zhu, 2024), knitr (v1.50; Xie, 2014, 2015, 2025b), lmtest (v0.9.40; Zeileis and Hothorn, 2002), pacman (v0.5.1; Rinker and Kurkiewicz, 2018), parallel (v4.4.3; R Core Team, 2025), performance (v0.15.0; Lüdtke et al., 2021), Polychrome (v1.5.4; Coombes et al., 2019), posterior (v1.6.1; Bürkner et al., 2025), psych (v2.5.6; Revelle, 2025), pwr (v1.3.0; Champely, 2020), random (v0.2.6; Eddebuettel, 2017), readr (v2.1.5; Wickham et al., 2024a), rmarkdown (v2.29; Allaire et al., 2024; Xie et al., 2018, 2020), rQCC (v2.22.12; Park et al., 2022; Park and Wang, 2020, 2022), rstan (v2.32.7; Stan Development Team, 2025a), rstudioapi (v0.17.1; Ushey et al., 2024), scrutiny (v0.5.0; Jung, 2024), stringr (v1.5.1; Wickham, 2023), tibble (v3.3.0; Müller and Wickham, 2025), tidybayes (v3.0.7; Kay, 2024c), tidyr (v1.3.1; Wickham et al., 2024b), tidyverse (v2.0.0; Wickham et al., 2019), tikzDevice (v0.12.6; Sharpsteen and Bracken, 2023), trialr (v0.1.6; Brock, 2023), and weights (v1.1.2; Pasek et al., 2025).

8.2 BKMO function

A simple implementation of a Bayesian Kaiser-Meyer-Olkin (BKMO) index function in R (v4.4.3 [2025-02-28 ucrt]; R Core Team, 2025), which computes the posterior 'sampling adequacy' from the posterior inter-correlations estimated by a model fit with the brms R package (v2.22.0; Bürkner, 2017, 2018), is provided in the code below:

```
1 # Compute BKMO index from a BGLM
2 #
3 # @param r A n by k matrix of estimated inter-correlations
4 # @return A n by p matrix of BKMO indices
```

```

5
6 BKMO <- function(r = NULL){
7   if(!is.matrix(r)) r <- as.matrix(r)
8
9   # Compute p (i.e., number of theorized manifestations)
10  disc <- sqrt(1 + 8 * ncol(r))
11  p <- (1 + disc) / 2
12
13  # Define utility function for vectorized operation
14  utility_KMO <- function(r = r, p = p){
15    # Construct correlation matrix
16    R <- diag(1, p)
17    R[upper.tri(R)] <- r
18    R[lower.tri(R)] <- t(R)[lower.tri(R)]
19    row.names(R) <- paste0('m', 1:p)
20    colnames(R) <- paste0('m', 1:p)
21
22    # Compute inverse R
23    R_inv <- solve(R)
24
25    # Compute anti-image R
26    R_q <- -R_inv / sqrt(outer(diag(R_inv), diag(R_inv)))
27
28    # Compute KMO
29    r2 <- R^2
30    q2 <- R_q^2
31    diag(r2) <- 0
32    diag(q2) <- 0
33    kmo <- c(
34      rowSums(r2) / (rowSums(r2) + rowSums(q2)),
35      overall = sum(r2) / (sum(r2) + sum(q2))
36    )
37    return(kmo)
38  }
39
40  # Compute KMO for each posterior draw using vectorization
41  bkmo <- do.call(rbind, lapply(1:nrow(r), function(i) utility_KMO(
42    r = r[i, , drop = FALSE], p = p
43  )
44  )
45  )
46  return(as.data.frame(bkmo))

```

47 }

```
1 ## Example
2
3 # Load packages
4 library("tidyverse")
5 library("EFA.dimensions")
6
7 # Load the Rosenberg (1965) data from EFA.dimensions
8 data(data_RSE)
9 df <- data_RSE
10 colnames(df) <- paste0("m", 1:ncol(df))
11
12 # Define maximum-likelihood functions
13 sd2 <- function(x = NULL, na.rm = FALSE){
14   stopifnot(is.numeric(x))
15   stopifnot(is.logical(na.rm))
16
17   x_mean <- mean(x, na.rm = na.rm)
18   x_var <- mean((x - x_mean)^2, na.rm = na.rm)
19   x_sd <- sqrt(x_var)
20   return(x_sd)
21 }
22
23 standardize2 <- function(x = NULL, na.rm = FALSE){
24   stopifnot(is.numeric(x))
25   stopifnot(is.logical(na.rm))
26
27   x_mean <- mean(x, na.rm = na.rm)
28   x_var <- mean((x - x_mean)^2, na.rm = na.rm)
29   x_sd <- sqrt(x_var)
30
31   x_standardized <- (x - x_mean) / x_sd
32   return(x_standardized)
33 }
34
35 # Standardize data matrix
36
37 df <- apply(df, 2, standardize2) %>% as.data.frame()
38
39 # Specify MCMC
40 library("parallel")
41 posterior_samples <- 4000
42 chains <- cores <- ifelse(parallel::detectCores()>1, parallel::detectCores()-1,
43   ↪ parallel::detectCores())
```

```
41 warmup <- 1000
42 iter <- ceiling((posterior_samples / chains) + warmup)
43 posterior_samples <- (iter - warmup)*chains
44 SEED <- 89881187
45 set.seed(SEED)
46
47 # Specify Bayesian Generalized Linear Model (BGLM)
48 library("brms")
49 bglm_model <- brms::bf(
50   mvbind(m1, m2, m3, m4, m5, m6, m7, m8, m9, m10) ~ 0,
51   sigma ~ 0,
52   family = gaussian(
53     link = "identity"
54   )
55 ) + set_rescor(TRUE)
56
57 # Specify 'weakly-informative' model priors
58 bglm_priors <- brms::get_prior(bglm_model, data = df)
59 bglm_priors[1,] <- brms::set_prior(
60   "lkj(2)",
61   class = "rescor"
62 )
63
64 # Fit Bayesian multivariate linear model with brms
65 bglm_fit <- brms::brm(
66   formula = bglm_model,
67   family = gaussian(
68     link = "identity"
69   ),
70   prior = bglm_priors,
71   data = df,
72   chains = chains,
73   cores = cores,
74   iter = iter,
75   warmup = warmup,
76   seed = SEED
77 )
78
79 # Extract posterior draws of the (residual) correlations from the fitted Bayesian
80   ↪ model
81 library("stringr")
82 bglm_samples <- brms::as_draws_df(bglm_fit)
```

```

82 bglm_correlations <- bglm_samples[, stringr::str_detect(colnames(bglm_samples), "
    ↪ rescor_")]
83
84 # Compute Bayesian KMO index
85 library("bayestestR")
86 kmo_bayes <- BKMO(r = bglm_correlations)
87 hist(kmo_bayes$overall)
88 print(
89   paste0(
90     "Mean = ", mean(kmo_bayes$overall) %>% round(digits = 3),
91     " (SD = ", sd2(kmo_bayes$overall) %>% round(digits = 3),
92     "; 95% HDI[",
93     bayestestR::hdi(kmo_bayes$overall)$CI_low %>% round(digits = 3), "; ",
94     bayestestR::hdi(kmo_bayes$overall)$CI_high %>% round(digits = 3), "]"
95   )
96 )

```

8.3 Robust BKMO index

As briefly discussed in section 3.1, the BGLM used for the BKMO index relies on a multivariate Gaussian likelihood, which constitutes a modeling assumption that may be violated, such as in relation to non-linear data and outliers. Deriving a robust BKMO index is trivial and briefly discussed in this section.

The BKMO index can easily be made more robust against non-linearity and the presence of outliers by *rank-standardizing* the items before computing the posterior correlation matrix. This is due to that transformation changing the correlations from the product-moment correlation coefficient to the more robust Spearman's (1904; 1910) *rank-correlation coefficient* (Field, 2018: 351-352), which instead of linearity makes the less strict assumption of *monotonicity* (cf. Clapham and Nicholson, 2014).

Alternatively, the BKMO index can also be made more robust to outliers by changing the BGLM specification from relying on a multivariate Gaussian likelihood to a *multivariate Student t* ($\mathcal{MV}\mathcal{T}$; Gelman et al., 2014: 580), due to that distribution being more outlier-prone (Juárez and Steel, 2010; O'Hagan, 1979). This, however, requires manually editing the Stan code produced by brms (Bürkner, 2017, 2018), since such a model is unsupported by that package at the time of writing.